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B2 OVERHEAD LINES

PS2 - Health assessment and refurbishment of OHL

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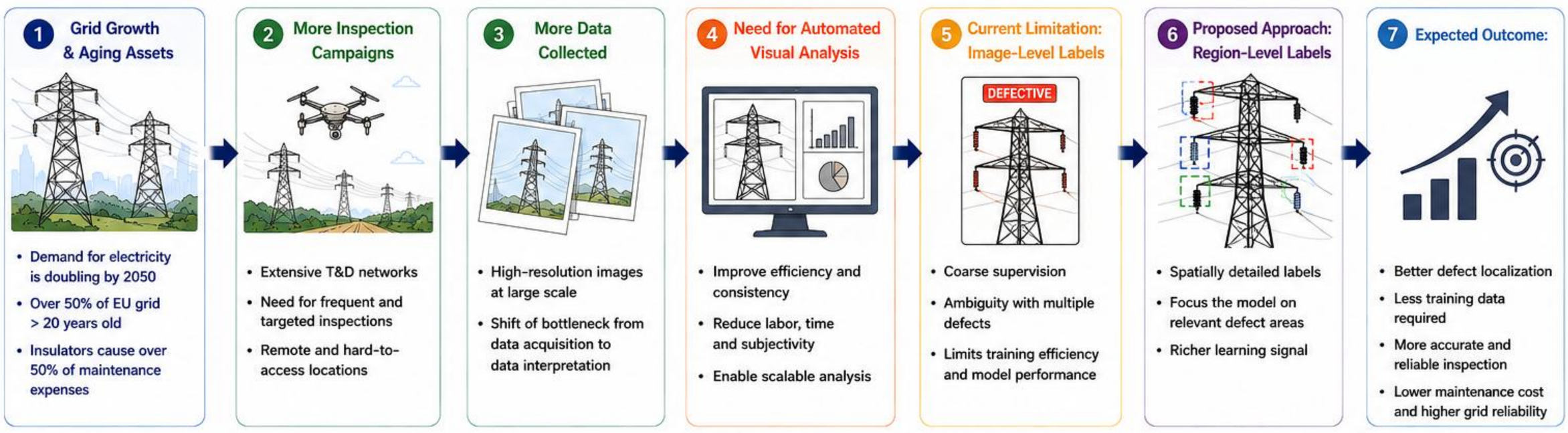
Automating Electrical Grid Asset Inspections: Improving Insulators Health Assessment with Region-Level Annotations

Pedro Daniel Rocha[1,2], André Coelho[4], **Ricardo Santos[4]**, Rui Martins[4], Luís Cruz[12], Fernando Lopes[1,3]

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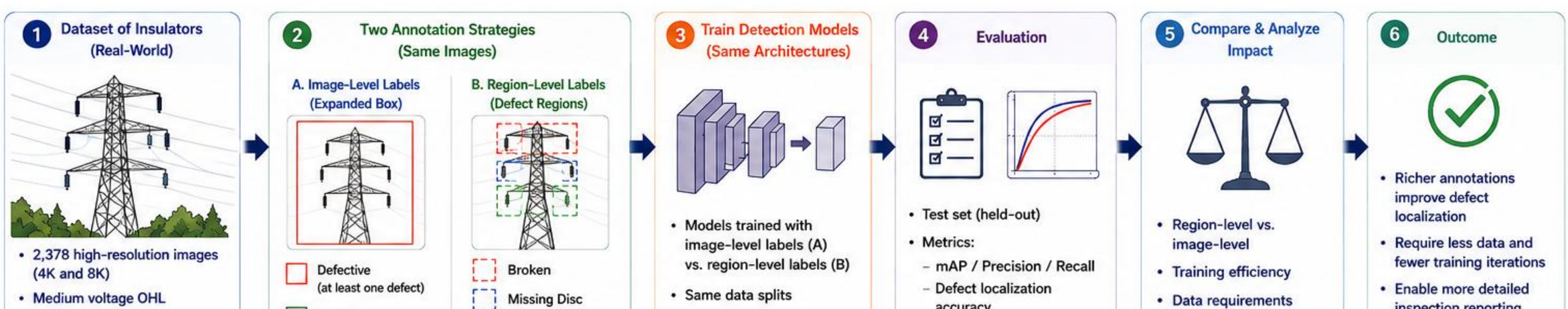
Context

- Electrical grid reliability is increasingly critical due to rising power demand and aging infrastructure.
- More than 50% of the EU grid has been in operation for over 20 years.
- Insulators represent only 5–8% of installation costs but account for over 50% of maintenance expenses.
- Manual visual inspections are labor-intensive, time-consuming, and subject to human variability.



Problem

- Automated inspection models are increasingly used to support maintenance decisions.
- Existing annotation workflows are designed for asset condition reporting rather than model training.
- Image-level labels provide only coarse supervision and no defect location information.
- Images containing multiple defects introduce ambiguity during training.
- The impact of richer spatial supervision on model learning remains unclear.



Objective

- Quantify the impact of annotation granularity on defect detection performance.
- Compare image-level and region-level supervision using identical images, architectures, and training procedures.
- Evaluate effects on accuracy, convergence, data efficiency, and robustness.

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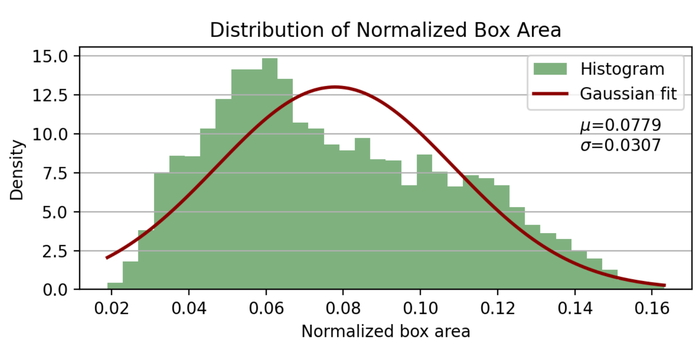


Methodology

- Two annotation strategies:
  - fitted bounding boxes
  - full-image "clamped" boxes
- Same NN Backbone - YOLOv11-m
  - 4-fold cross-validation
  - 3 random seeds

Dataset

- 4,756 images collected between 2023–2025 in rural and suburban environments
- 36,379 annotated ceramic and glass insulators (broken and non-defective)





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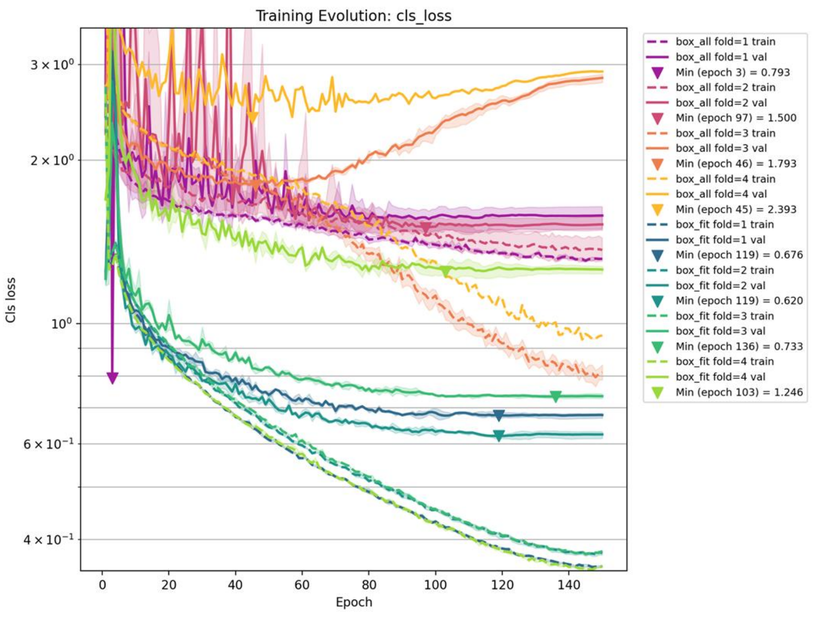
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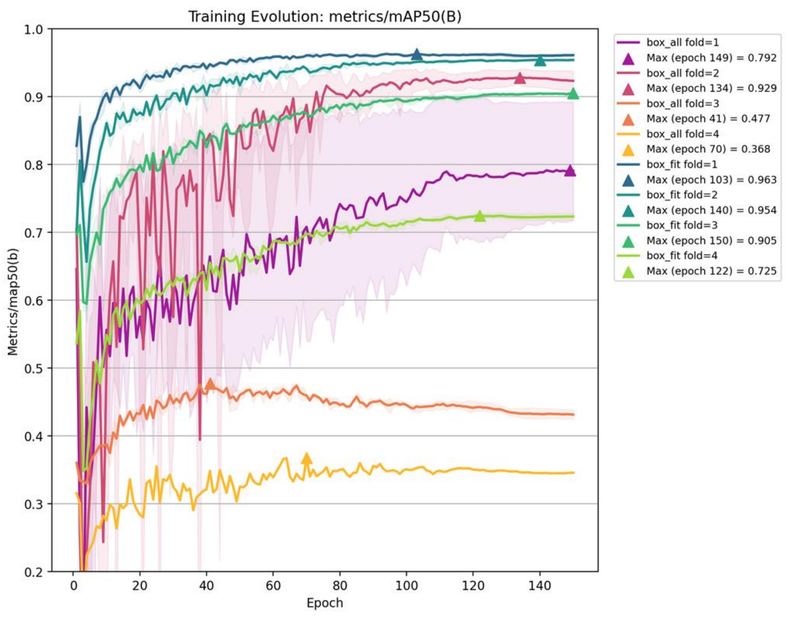
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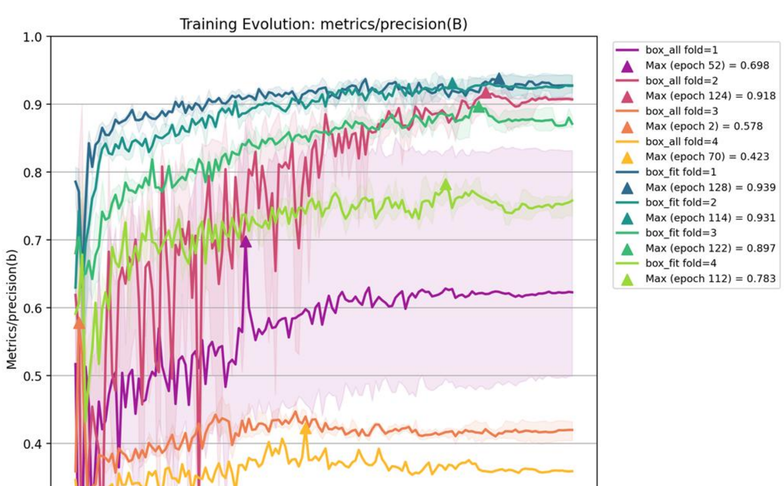
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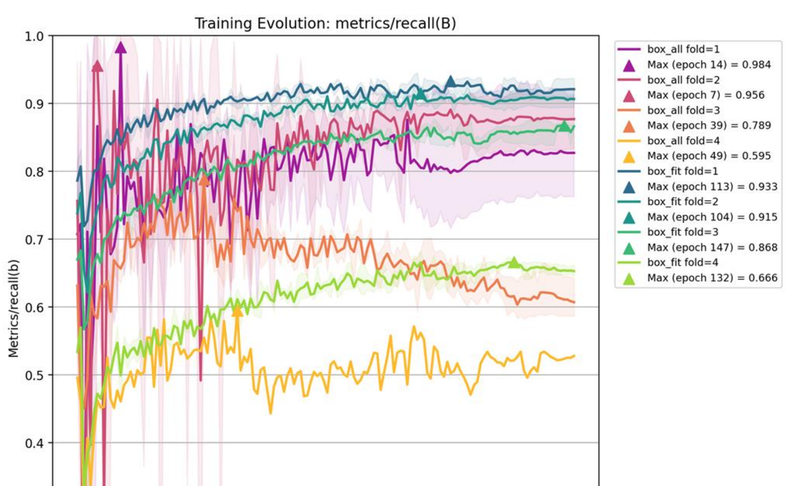
Key Findings

- Region-level supervision achieved higher accuracy and lower variability.
- Models converged faster and exhibited more stable optimization.
- Precision–recall performance improved consistently across folds.
- Image-level supervision was more susceptible to overfitting and false correlations.
- Benefits were strongest for larger, more visible assets.











Practical Implications

- Spatial supervision improves feature learning beyond localization alone.
- Richer annotations reduce data requirements and improve training stability.
- More reliable defect detection supports scalable automated inspections.
- Additional annotation effort is justified by performance gains.

Future Work

- Multi-defect scenarios
- Multimodal sensing
- Semi-supervised learning

Limitations

- Single defect type (broken insulators); multi-defect scenarios unexplored
- Visible-light imagery only—other modalities (thermal, LiDAR) untested
- Only YOLOv11-m evaluated; generalization to other architectures unclear

References

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